Tarrik Quneibi • (517) 206-8821 • tarrikq@gmail.com • Portfolio

EDUCATION

University of Michigan
Bachelor of Science in Environmental Engineering
Ann Arbor, Michigan
Graduation: December 2021

University of Michigan
Master of Science in Environmental Engineering
Ann Arbor, Michigan
Graduation: December 2022

WORK EXPERIENCE

ISS Corporate (SustainaBase)

Senior Data Engineer Feb 2023 - Present

- **Contributed to SustainaBase's successful acquisition by ISS Corporate** by serving as a key technical stakeholder during the due diligence process, participating in executive-level interviews and translating the platform's cloud-native AWS architecture and emissions reporting capabilities into demonstrable business value.
- **Architected and deployed cloud-native AWS Infrastructure-as-Code pipelines** with deep knowledge of environmental reporting standards (GHG Protocol, SASB, CDP, CSRD), leveraging a cross-disciplinary background in environmental and data engineering to deliver consistent, audit-ready emissions disclosures across development, staging, and production environments.
- **Designed and managed multi-tiered database architecture** integrating relational, NoSQL, and data lakehouse systems to support large-scale environmental data storage, versioning, enrichment, and rapid querying. Enabling both internal analytics and external client reporting across multiple greenhouse gas emission categories. Reducing processing cycles from weekly to seconds, accelerating client reporting turnaround.

University of Michigan

Research Assistant / Project Lead Jan 2022 - Feb 2023

- **Led and managed a multidisciplinary city-wide water quality research project** end-to-end - coordinating scheduling, field sampling logistics, and task delegation across municipal water treatment facility managers, local businesses, and university research staff - to evaluate UV disinfection's effect on opportunistic pathogens in drinking water distribution systems and present technical findings to diverse stakeholders.
- **Conducted comprehensive water quality assessments** including physicochemical characterization, biological sampling, and operational data collection (UV contact time, treatment parameters) across multiple distribution system locations, directly supporting water risk evaluation.
- **Established reproducible data management workflows** integrating field measurements, laboratory analysis, and treatment plant operational data into standardized formats with API-driven pipelines, enabling consistent reporting and seamless data updates across the project team.

City of Ann Arbor Drinking Water Treatment Plant

Water Resource Engineering Intern May 2021 - Jan 2022

- **Designed and executed an instrumentation validation study** for newly installed post-filtration particle counters, developing controlled experiments using calibrated particle suspensions to verify detection reliability for *Cryptosporidium*-sized oocysts and ensure compliance with Surface Water Treatment Rule requirements.
- **Built time-series data processing pipelines** in R to ingest, clean, and analyze continuous particle count streams, applying statistical methods to correlate filtration performance indicators, turbidity, flow rate, and runtime, with pathogen-surrogate particle breakthrough events.
- **Developed an operational decision-support dashboard** integrating real-time particle counter data with disinfection contact time parameters, enabling plant operators to make informed, data-driven treatment adjustments and supporting CT compliance verification across contactor zones.

TECHNICAL SKILLS

Programming & Tools: Python, R, SQL, Git/GitHub, AWS (Lambda, S3, Step Functions, Glue, Athena, CDK), Java Spring Boot, JavaScript	**Data & Databases:** SQL Server, DynamoDB, MySQL, S3 Data lakehouse architecture, Database management principles, Data Governance	**Domain Expertise:** Geospatial analysis (Google Earth Engine, GIS), Water quality & hydrologic modeling, Sustainability Reporting & Disclosure Environmental data engineering,

PROJECT WORK

Event-Driven Cloud Architecture for Real-Time Environmental Data Processing

- Designed serverless event-driven architecture using AWS EventBridge for orchestration, routing data from multiple ingestion sources (S3 uploads, API Gateway, webhooks) to appropriate processing workflows, with Step Functions coordinating distributed Lambda execution for validation, transformation, emission factor matching, and GHG calculations across different emission categories. Producing audit-ready reports and dashboards aligned with GHG Protocol, SASB, and other reporting requirements.
- Engineered intelligent emission factor matching system using DynamoDB feature library with hierarchical lookup strategies and fuzzy matching algorithms, automatically enriching activity data with region-based and category-specific emission factors while delivering real-time analytics and status updates to users through event-driven dashboard updates - ensuring accurate, traceable emission calculations for external client reporting and auditing.
- Scaled platform from manual Excel-based processing of fewer than 20 clients to automated near-unlimited capacity through parallel event-driven architecture with asynchronous messaging (SNS/SQS), serverless auto-scaling, and automated security validation. Reducing processing cycles from weeks to seconds while maintaining complete data lineage, quality controls, and standardized reporting outputs.

Predictive Modeling: Harmful Algal Bloom Prediction Using Machine Learning

- Developed R-based data acquisition and processing workflows for large-scale water quality datasets from Lake Erie monitoring buoys, implementing reproducible pipelines for hydrologic data management and analysis-ready dataset preparation.
- Applied statistical methods (Spearman correlation, multiple imputation) and machine learning techniques (random forest feature selection) to quantify relationships between physicochemical water quality parameters and cyanotoxin concentrations, supporting evidence-based water risk evaluation.
- Built and trained deep learning neural network achieving 89% prediction accuracy for harmful algal bloom concentrations, demonstrating applied capabilities in water quality modeling, predictive analytics, and synthesis of technical findings for research reporting.

Geospatial Analysis and Modeling of Air Quality Data

- Implemented R-based geoprocessing workflows to transform tabular environmental monitoring data into spatial matrices, producing interactive geospatial visualizations including density plots and 3D surface maps to support technical communication of complex environmental datasets.
- Conducted spatial analysis and large-scale environmental monitoring of ground-level ozone concentrations across California using modern geospatial R libraries and Google Earth Engine, demonstrating proficiency in watershed and regional-scale environmental assessment.
- Developed gradient boosted decision tree models for air quality prediction achieving 90% accuracy, producing analysis-ready data products and demonstrating applied expertise in environmental modeling and computational environmental analysis.